

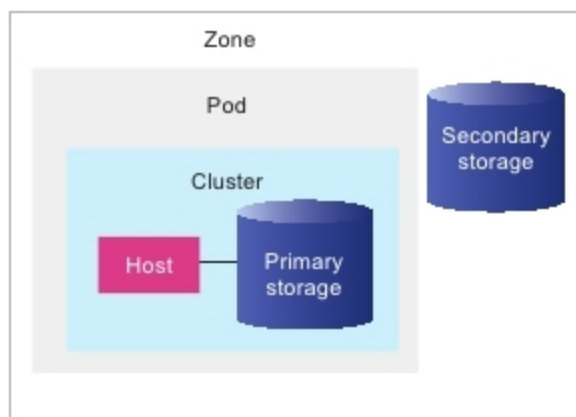


Figure 1: CloudStack architecture

a-Service CMP, which facilitates creating, deploying and managing cloud services by providing a complete stack of features and components for cloud environments. It uses existing hypervisors such as KVM, VMware vSphere, VMware ESXi, VMware vCenter and XenServer/XCP for virtualisation. CloudStack can also orchestrate the non-technical elements of service delivery such as billing and metering. It presents a range of APIs, allowing it to be integrated with any other platform.

The main components of CloudStack are:

- Compute nodes (CNs), which are servers when VM instances are instantiated.
- A cluster, which is composed of several CNs that share the same hypervisor and primary storage system.
- Pod is a rack of hardware including Layer-2 switches and one or more clusters. It is responsible for storing the VM files, which represent the primary storage.
- The Availability Zone is made up of one or more pods, with secondary storage forming a zone. This is equivalent to a single data centre, representing geographic allocations. The secondary storage stores VM templates, ISO images and disk volume snapshots.
- The management server is a single point of configuration that provides Web user interfaces and APIs, and manages the assignment of VM instances to particular hosts and of

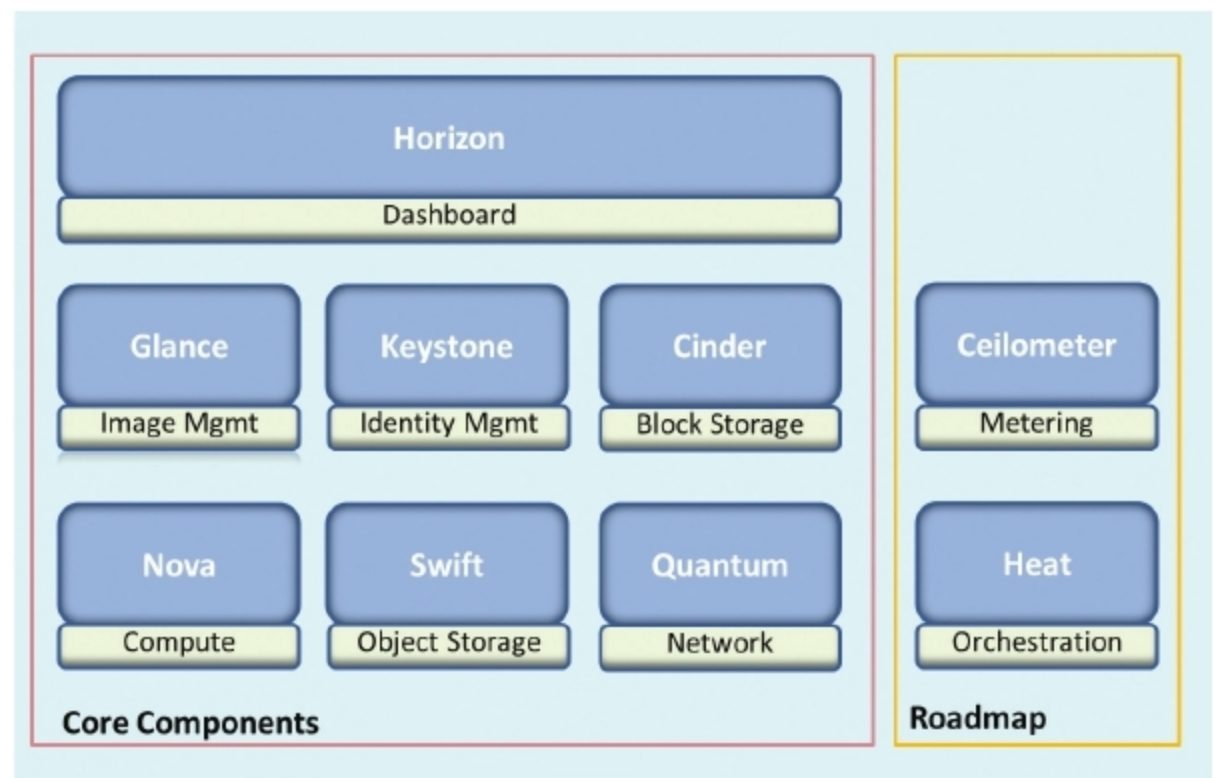


Figure 2: OpenStack components

public and private IP addresses to particular accounts, as well as the allocation of storage.

Features

- **Self-service user interface:** AJAX console access, multi-role support, network virtualisation, hypervisor agnostic, usage metering, virtual routers.
- **LVM support:** Block storage volumes, NetScaler support, OpenStack Swift integration, LDAP integration, domains and delegated administration.

Official website: <https://cloudstack.apache.org/>

Latest version: 4.12.0.0

OpenStack

OpenStack consists of a set of software tools for building and managing cloud computing platforms for public and private clouds using pooled virtual resources. The tools comprising the OpenStack platform are called projects. They handle core cloud computing services of compute, networking, storage, identity and image services. OpenStack software controls large pools of compute, storage and networking resources throughout a data centre, and

is managed through a dashboard or via the OpenStack API.

OpenStack consists of the following nine components.

- **Nova:** This is the primary computing engine used for deploying and managing a large number of virtual machines and instances to handle computing tasks.
- **Swift:** This is a storage system for objects and files.
- **Cinder:** This is a persistent block storage component for compute instances.
- **Neutron:** This provides networking capability so that all components can communicate quickly.
- **Horizon:** This is a GUI interface for OpenStack.
- **Keystone:** This provides identity services for OpenStack.
- **Glance:** This provides image services and allows images (virtual copies of hard disks) to be used as templates for deploying new virtual machine instances.
- **Ceilometer:** This provides telemetry services, and billing services to individuals.
- **Heat:** This is an orchestration component that allows developers to store the requirements of cloud applications in files.